A NOTE ON THE NESTED RADICAL ITERATION $x_{n+1} = \sqrt{2 + x_n}$

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ABSTRACT. We give a self-contained treatment of the iteration $x_{n+1} = \sqrt{2 + x_n}$ started at an arbitrary admissible point $a \geq -2$. We show that the iterates always converge to 2, with a geometric rate, and deduce the corresponding statement for the Cesàro means.

1. Introduction

The nested radical $\sqrt{2 + \sqrt{2 + \sqrt{2 + \cdots}}}$ is a standard first example of a recursively defined sequence. In most textbooks the iteration is started at $x_0 = 0$ or $x_0 = \sqrt{2}$, and convergence is obtained from the monotone convergence theorem. The purpose of this note is to record the argument for every starting value at which the iteration makes sense, that is, for every $a \geq -2$, and to make the elementary estimates explicit. Our main result is Theorem 2.8; in Section 3 we derive a statement about averages from a classical theorem of Cauchy.

Throughout, $\mathbb{N} = \{0, 1, 2, \dots\}$, and $\sqrt{\cdot}$ denotes the non-negative square root on $[0, \infty)$.

2. The iteration

Definition 2.1. Let $f\colon [-2, \infty) \to [0, \infty)$, $f(x) = \sqrt{2 + x}$. For $a \geq -2$ we define the sequence $(x_n)_{n \in \mathbb{N}} = (x_n(a))_{n \in \mathbb{N}}$ by $x_0 = a$ and $x_{n+1} = f(x_n)$ for $n \in \mathbb{N}$.

Since f takes values in $[0, \infty) \subset [-2, \infty)$, the sequence is well defined, and $x_n \geq 0$ for all $n \geq 1$.

Lemma 2.2. *For $x \geq -2$ we have $f(x) = x$ if and only if $x = 2$.*

Proof. Clearly $f(2) = \sqrt{4} = 2$. Conversely, if $f(x) = x$ then $x \geq 0$, since f is non-negative, and squaring gives $x^2 = x + 2$, i.e. $(x - 2)(x + 1) = 0$. As $x \geq 0$, we get $x = 2$. □

Lemma 2.3. *For every $x \geq -2$,*

$$|f(x) - 2| \leq \tfrac{1}{4} |x - 2|.$$

Proof. Rationalising,

$$f(x) - 2 = \frac{(2 + x) - 4}{\sqrt{2 + x} + 2} = \frac{x - 2}{\sqrt{2 + x} + 2}.$$

The denominator is at least 2, whence the claim. □

Lemma 2.4. *If $-2 \le x \le 2$, then $x \le f(x) \le 2$. If $x \ge 2$, then $2 \le f(x) \le x$.*

Proof. Let $-2 \le x \le 2$. Then $2 + x \le 4$, so $f(x) \le 2$. Since both sides are non-negative, the inequality $x \le \sqrt{2+x}$ is equivalent to $x^2 \le 2 + x$, that is, to $(x-2)(x+1) \le 0$, and this holds for $-2 \le x \le 2$.

Now let $x \ge 2$. Then $2 + x \ge 4$, so $f(x) \ge 2$; moreover $x \ge 2 > 0$, so $f(x) \le x$ is equivalent to $2 + x \le x^2$, i.e. to $(x-2)(x+1) \ge 0$, which holds. □

Corollary 2.5. *Let $a \ge -2$ and $x_n = x_n(a)$.*

(1) *If $a \le 2$, then $x_n \le x_{n+1} \le 2$ for all $n \in \mathbb{N}$.*
(2) *If $a \ge 2$, then $2 \le x_{n+1} \le x_n$ for all $n \in \mathbb{N}$.*

Proof. For (1), $x_0 = a \in [-2, 2]$, and if $x_n \in [-2, 2]$ then Lemma 2.4 gives $x_n \le x_{n+1} \le 2$; in particular $x_{n+1} \in [-2, 2]$. Induction on n finishes the proof. Part (2) is identical, using the second half of Lemma 2.4. □

Lemma 2.6. *Let $a \ge -2$. Then $|x_n(a) - 2| \le 2^{-n}|a - 2|$ for every $n \in \mathbb{N}$.*

Proof. For $n = 0$ there is nothing to prove. If the inequality holds for n, then, as $x_n \ge -2$, Lemma 2.3 gives $|x_{n+1} - 2| = |f(x_n) - 2| \le \frac{1}{2}|x_n - 2| \le 2^{-(n+1)}|a - 2|$. □

Lemma 2.7. *Let $a \ge -2$ and suppose that $x_n(a) \to L$ for some $L \in \mathbb{R}$ with $L \ne 2$. Then $L = -1$.*

Proof. Since $x_n \ge -2$ for all n, $L \ge -2$, and f is continuous on $[-2, \infty)$. Letting $n \to \infty$ in $x_{n+1} = f(x_n)$ gives $L = \sqrt{2 + L}$, hence $L^2 = L + 2$ and $L \in \{2, -1\}$. As $L \ne 2$, $L = -1$. □

Theorem 2.8. *For every $a \ge -2$ the sequence $(x_n(a))_{n\in\mathbb{N}}$ converges, and*

$$\lim_{n\to\infty} x_n(a) = 2.$$

Proof. If $a \le 2$, then by Corollary 2.5(1) the sequence is non-decreasing and bounded above by 2; if $a \ge 2$, it is non-increasing and bounded below by 2 by Corollary 2.5(2). In either case the monotone convergence theorem shows that $x_n \to L$ for some $L \ge -2$. Passing to the limit in $x_{n+1} = f(x_n)$, using continuity of f at L, gives $f(L) = L$, so $L = 2$ by Lemma 2.2. (Alternatively, if $L \ne 2$ then $L = -1$ by Lemma 2.7, which is impossible because $x_n \ge 0$ for $n \ge 1$ forces $L \ge 0$.) Lemma 2.6 gives the same conclusion directly, together with a geometric rate. □

3. Cesàro means

Definition 3.1. For a real sequence $y = (y_n)_{n\in\mathbb{N}}$ and $N \ge 1$, the N-th Cesàro mean of y, i.e. the arithmetic mean of its first N terms, is

$$\sigma_N(y) = \frac{1}{N} \sum_{n=0}^{N} y_n.$$

The following is the classical theorem of Cauchy on the regularity of the Cesàro method; see [1] and [2, Chapter V].

Theorem 3.2 (Cauchy)**.** *Let $(y_n)_{n\in\mathbb{N}}$ be a real sequence and $L \in \mathbb{R}$. Then $y_n \to L$ if and only if $\sigma_N(y) \to L$ as $N \to \infty$.*

Corollary 3.3. *Let $0 \le a \le 2$ and $x_n = x_n(a)$. Then $\sigma_N(x) \to 2$ as $N \to \infty$.*

Proof. By Theorem 2.8, $x_n \to 2$. Theorem 3.2 now gives $\sigma_N(x) \to 2$. □

References

[1] A.-L. Cauchy, *Cours d'analyse de l'École royale polytechnique. Ire partie: Analyse algébrique*, Debure, Paris, 1821.
[2] G. H. Hardy, *Divergent Series*, Clarendon Press, Oxford, 1949.

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